

Interval Prediction of Aeroengine Remaining Useful Life Based on BCT-Bayesian Bidirectional LSTM Network

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Considering the uncertainty due to the inconsistency of data disturbance and the variation of operation condition, the accurate estimation of remaining useful life (RUL) is challenging. This article proposes an interval prediction method based on box-cox transformation-Bayesian bidirectional long and short time memory (BCT-Bayesian-biLSTM) network. The proposed method has two highlights to improve the prediction accuracy. First, BCT is presented to make the different data distribution consistent with the required hypothesis distribution which improves the model uncertainty. Then, a probabilistic model of bidirectional long and short time memory network based on Bayesian deep learning framework is established to extract the temporal correlation features hidden in contextual data. The parameters of BCT-Bayesian-biLSTM are expressed as random variables to realize the RUL point prediction and its interval estimation simultaneously. Comparison experiment results with the state-of-art methods show that the proposed

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method has the smallest prediction error and a narrower prediction interval. It exhibits the advantage of the prediction accuracy and the model generalization ability for unknown distribution data.

I. INTRODUCTION

The performance state of aircraft engine directly affects the overall safety and reliability of aircraft and relevant flight personnel [1], [2]. The accurate remaining useful life (RUL) prediction enables airlines to optimize maintenance schedules, reduce unexpected downtime and prevent potential in-flight failures [3]. It not only enhances safety but also leads to significant cost savings by avoiding unnecessary replacements and maximizing engine utilization. With the information technology advances, sophisticated predictive models incorporating real-time data and artificial intelligent technology are becoming increasingly vital for maintaining the delicate balance between safety, performance, and economic viability in the highly competitive aerospace industry [4].

Bayesian neural network (BNN) is a popular structure for the uncertainty prediction which shows good stability and accuracy on many datasets of complex equipment systems [5]. A Bayesian deep learning prediction framework is proposed for RUL uncertainty quantification and calibration [6]. The accidental uncertainty caused by measurement errors and cognitive uncertainty caused by model errors are integrated to quantify the prediction results. For the probabilistic RUL prediction problem, Pan et al. [7] proposed a fusion method of Bayesian learning and stochastic process. Nguyen et al. [8] combined the deep recurrent neural network (RNN) and probabilistic model to predict the RUL distributions. The idea of Bayesian learning are integrated into bidirectional RNN to characterize the uncertainty of the predicted degradation level via the variational inference (VI) technique [9]. Under the Bayesian learning framework, the calculated amount of model solving increases significantly since the model parameters are random variables. The computational cost problem should be considered during the Bayesian learning. The dropout-based Bayesian approximation technique is proposed to reduce the model complexity, but the computing in multiple repeated sampling are still required [10]. Although Bayesian learning provides a possibility framework to deal with uncertainty modeling problems, it is often based on some assumptions, such as the appropriate priors for model parameters, the data with independent and stationary distribution function. These assumptions may not hold for the dynamic, time-series, or spatially correlated data.

Considering the data have the temporal correlation characteristics and inconsistent distribution with the assumption, this article proposes an interval prediction method based on the Bayesian bidirectional long and short time memory (biLSTM) network with box-cox transformation (BCT). It tackles the uncertainty issues in RUL prediction through the following highlights: An interval prediction effectively to quantify the multiple uncertainties, the temporal correlation feature recognition based on a biLSTM

network, and a distribution transformation mechanism to solve the inconsistency between actual distribution and the assumption.

The main contributions include the following.

- 1) An interval prediction method named as BCT-Bayesian-biLSTM is proposed based on the bidirectional LSTM network under the framework of Bayesian deep learning. It quantifies the multiple uncertainties caused by measurement error and cognitive learning.
- 2) Bi-LSTM captures the richer temporal correlation features, and BCT transforms the real data with different distribution into the standard gaussian consistent with the preassumption. The highlight focuses on the exploring of the time sequence and uncertainty.
- 3) The aeroengine RUL is predicted based on the proposed BCT-Bayesian-biLSTM model. Its superiority is shown at its higher accuracy and the statistical stability of prediction result even facing with different data distribution.

The rest of this article is organized as follows. Section II gives the related work to show the motivation and the advantage of the proposed method. Section III analyzes the uncertainty sources of RUL prediction and gives the methodology first, then establishes the application framework in the aero engine residual life prediction problem. Section IV shows the experiment and discussion. Finally, Section V concludes this article.

II. RELATED WORK

The prediction methods of RUL are mainly divided into two categories: 1) physics-based; and 2) data-driven [11]. Physics-based prediction is to establish a mathematical model according to the failure and performance degradation mechanism, such as Wiener, Gaussian, and Markov models [12], [13]. While physics-based methods provide a solid theoretical foundation, they typically require detailed equipment modeling and extensive prior knowledge. Their predictive accuracy and reliability are often limited when applied to complex systems or in the absence of precise models. The data-driven approaches include statistical analyzing, deep learning, and machine learning which show the advantage at powerful feature learning and nonlinear modeling capabilities [14], [15]. In these data-driven methods, Bayesian networks particularly exhibit strong adaptability and flexibility in capturing complex degradation patterns, while also enabling uncertainty quantification through interval-based predictions. Now, the hybrid RUL prediction is popular by combining any of the former approaches, such as the empirical knowledge and data learning techniques [13], [16], [17].

Deep learning has achieved remarkable advances in RUL prediction. An ACDBN-SSA-KELM method is proposed which exhibits fast convergence and high computational efficiency [18]. However, it is primarily

applicable to relatively simple rolling bearing systems. Graph convolutional networks (GCNs) are adopted to model the dependencies among multisensors [19], [20]. In particular, the gated GCN [20] enables the uncertainty quantification, but it heavily relies on the high-quality data and substantial computational resources. Moreover, transformer-based architectures have also been explored for RUL prediction. For example, Mo et al. [21] integrated gated convolutional units (GCUs) to extract multiscale features, and Zhang et al. [22] enhanced the temporal modeling capability through a dual aspect self-attention mechanism. Although these approaches demonstrate superior performance, they tend to be structurally complex and difficult to fine-tune.

The above researches design a deterministic prediction model which provides a point estimation of RUL, and typically focusing on how to improve the prediction accuracy. However the data collected by multisensors are often interfered by uncontrollable disturbances, and are not possible to cover all operating conditions. These causes lead to the cognitive uncertainty of data-driven prediction model. Interval prediction is more capable of quantifying predictive uncertainty under nonstationary conditions and data distribution shifts. In contrast, point prediction provides only a single estimated value which ignores the effect from inherent noise and uncertainty in both modeling and data acquisition. Point prediction may pose the overconfidence and potential safety risks in practical applications. Interval prediction aligns better with the safety standards and the decision-making logic of industrial maintenance. It facilitates more scientific planning of maintenance windows and spare parts inventory, and avoid the premature maintenance that wastes resources or delayed maintenance that leads to catastrophic failures. Therefore, it is important to study the cognitive uncertainty under the framework of data-driven learning, i.e., the interval prediction or estimation.

The proposed BCT-Bayesian-biLSTM interval prediction model first incorporates the BCT module, which effectively addresses the challenges of the long-tail effect and mismatch of data distribution in real sensor data. The design of Bi-LSTM structure considers both forward and backward information in the sequence data, and helps to capture the richer temporal correlation features. The use of Bayesian learning strategy enables the effective modeling of uncertainty in the prediction process. So the proposed method enhances the accuracy and consistency of the prediction results, and it also shows good model adaptability when the actual application encounters different distributions. Compared to traditional physical-based prediction methods, it overcomes the limitations of imprecision inherent when physically modeling the complex systems. It also avoids the reliance of purely data-driven methods on large volumes of high-quality historical data and labels.

III. BCT-BAYESIAN-BILSTM-BASED PREDICTION

The engine system with a complex structure is susceptible to a variety of uncertain factors, such as noise and disturbance from the different operation environments and

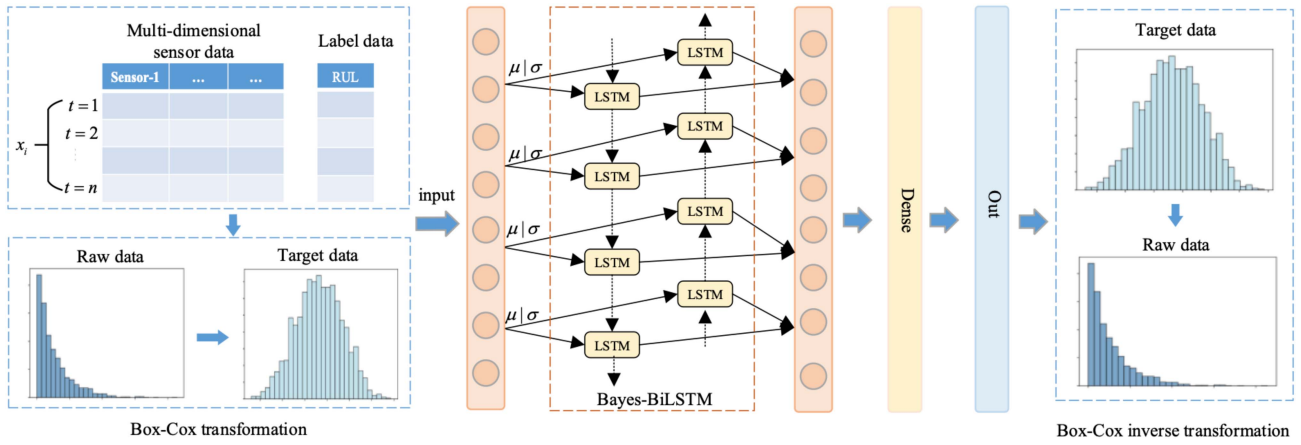


Fig. 1. Overall framework of BCT-Bayes-BiLSTM. It consists of three main components: Box-cox transformation, Bayesian-BiLSTM network and box-cox inverse transformation.

conditions, and it will results the randomness of sampling data. Moreover, the inaccuracy and poor adaptability of data learning models also lead to the cognitive uncertainty of prediction results [23]. To assure the stability and safety of turbofan engines, it is necessary to further consider a variety of uncertain factors and integrate them into the RUL prediction model. So the interval prediction is important for formulating more sensible maintenance strategies.

We propose a BCT-Bayesian-biLSTM network to solve the accurate interval prediction problem, and its framework is shown in Fig. 1. A probabilistic prediction model of the biLSTM network with the Bayesian deep learning strategy is established. Various uncertainties from learning model and sampling data are modeled through Bayesian inference to quantify the prediction uncertainty or interval. Here, the Bayesian deep learning strategy introduces a prior distribution of model parameters which are regarded as random variables. The likelihood loss of data samples is established to optimize the posterior probability of model parameters. The proposed method expresses the modeling uncertainty with a form of probability distribution. The Bayesian deep learning strategy guarantees the proposed network with good uncertainty modeling ability, high adaptiveness and generalization.

However, the traditional Bayesian learning algorithm ignores the temporal correlation, and has strict hypothesis on data distribution, which leads to the limitation in practical applications. So this article first proposes a box-cox transform mechanism to guarantee the distribution of real data is consistent with the assumed distribution for learning. The Bi-LSTM network structure is introduced to capture the sequential features hidden in the time series. The proposed method highlights at three terms to enhance the prediction accuracy, Bayesian learning-based uncertainty modeling, BCT for distribution consistence, and BiLSTM for sequential feature extraction.

A. BCT Mechanism

The Bayesian deep learning strategy is common to assume a specific prior distribution such as the normal

distribution, then optimizes the model by constantly updating the posterior distribution with observational data. The model accuracy is related with the distribution of model parameters and input data. When the distribution of real data is inconsistent with the presupposed one, the model is hard to grasp the relationship between input and output [24]. In order to enhance the prediction accuracy of the Bayesian deep learning model, it is necessary to transform the original data to assure its distribution approximates the assumed distribution. This transformation would reduce the impact of uncertainty caused by distribution inconsistency.

Usually a Gaussian normal distribution $N(\mu, \sigma^2)$ is presupposed to describe the statistical characteristics of data. Here, μ and σ represent the sample mean and standard deviation, respectively. BCT is commonly in statistics to convert the continuous data with different distribution into the normal distribution state by introducing a transformation parameter λ . The normality, linearity, independence, and homogeneity of variance are effectively improved without losing the original data information. The general form of BCT is expressed as

$$y_\lambda = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \lg y, & \lambda = 0 \end{cases} \quad (1)$$

where y and y_λ stands for the raw data and the transformed sequence, respectively. The corresponding inverse transformation is

$$y = \begin{cases} (\lambda y_\lambda + 1)^{1/\lambda}, & \lambda \neq 0 \\ \exp(y_\lambda), & \lambda = 0. \end{cases} \quad (2)$$

BCT can transform the original data y to y_λ which follows a normal distribution in which parameter λ is appropriately selected [25]. It is worth noting that BCT includes many special transformations, such as logarithmic transformation and reciprocal transformation. BCT is the natural logarithmic transformation if $\lambda = 0$, the square root transformation if $\lambda = 0.5$, and reciprocal transformation if $\lambda = -1$. Parameter λ is usually optimized by the maximum likelihood estimation.

Consider the original measured process data $X \in R^{n \times m}$ and the output RUL data $Y \in R^{n \times 1}$, where m and n are the number of independent variables and the samplings. The output sequence after BCT is $Y_\lambda = [Y_{1\lambda}, Y_{2\lambda}, \dots, Y_{n\lambda}]^T$. The linear regression of output variable is given as follows:

$$Y_\lambda = X\beta + \varepsilon \quad (3)$$

where $\beta = [\beta_0, \beta_1, \dots, \beta_m]$ is the regression coefficient, and random error $\varepsilon \sim N(0, \sigma^2 I_n)$, I_n is n -dimensional identity matrix. Consider the transformed output $Y_\lambda \sim N(X\beta, \sigma^2 I_n)$, and define the likelihood function as

$$L(\beta, \sigma^2, \lambda|Y) = \frac{1}{(\sqrt{2\pi}\sigma^2)^n} \exp\left(-\frac{1}{2\sigma^2} (Y_\lambda - X\beta)^T (Y_\lambda - X\beta)\right) \cdot J(\lambda, Y) \quad (4)$$

with the BCT Jaboci determinant $J(\lambda, Y) = \prod_i Y_i^{\lambda-1}$.

The log-likelihood function of (4) yields

$$\begin{aligned} \ln L = & -\frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \sigma^2 - \frac{[Y_\lambda - X\beta]^T [Y_\lambda - X\beta]}{2\sigma^2} \\ & + (\lambda - 1) \sum_{i=1}^n \ln Y_i. \end{aligned} \quad (5)$$

Under the fixed transformation parameter λ , the maximum likelihood estimations $(\hat{\beta}, \hat{\sigma}^2)$ for regression parameters could be obtained as

$$\begin{aligned} \hat{\beta}(\lambda) &= (X^T X)^{-1} X^T Y_\lambda \\ \hat{\sigma}^2(\lambda) &= \frac{1}{n} \cdot (Y_\lambda - X\beta)^T (Y_\lambda - X\beta). \end{aligned} \quad (6)$$

Substituting estimation (6) into (5)

$$\ln L_{\max}(\lambda|Y) = -\frac{n}{2} \ln \hat{\sigma}^2 + (\lambda - 1) \sum_{i=1}^n \ln Y_i + C \quad (7)$$

where constant $C = -n/2 \cdot \ln 2\pi - n/2$. For the given data, the maximum likelihood function $\ln L_{\max}$ is searched multidimensional, and the value at the extreme value is the optimal parameter λ .

The data transformation in the proposed BCT-Bayesian-biLSTM network is expressed as

$$Y_\lambda = f_{\text{BCT}}(X, Y). \quad (8)$$

The BCT model f_{BCT} meets the condition of invertibility, and the predicted output needs to be restored. Suppose the RUL prediction model is described as $f_{\text{pre}}(x^*, \theta) + \epsilon$ which are obtained from the Bayesian-biLSTM network. Suppose the test data x^* is fed into the biLSTM model, and the prediction result is denoted as y^* . Through the inverse BCT transformation f_{BCT}^{-1} , the RUL prediction is obtained in the original data space, namely

$$y_{\text{pre}} = f_{\text{BCT}}^{-1}[f_{\text{pre}}(x^*)]. \quad (9)$$

B. BiLSTM Network

Now, consider the RUL interval prediction model biLSTM network, whose hidden layer is implemented by two independent LSTM modules propagating in forward and

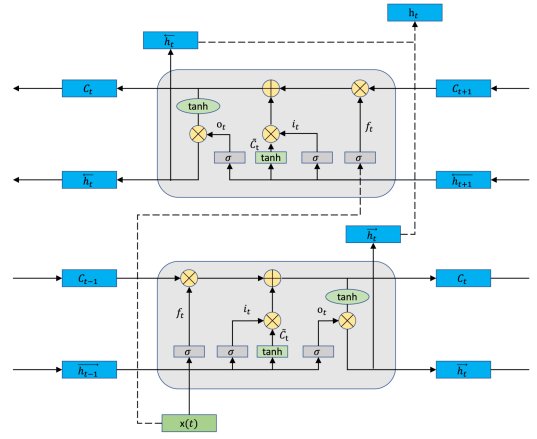


Fig. 2. BiLSTM. It has forward and backward sequences in opposite directions. The concatenated hidden states enhance temporal feature learning by capturing both past and future context.

backward directions. Its structure is shown in Fig. 2. It is known that LSTM is an improved RNN to solve the gradient disappearance and long-term dependence [26]. LSTM is a unidirectional structure to transmit information, so it affects the analysis ability for time series. BiLSTM introduces the bidirectional structure which simultaneously considers the influence of past and future states. It captures more dependency relationship before and after the series than the traditional LSTM and has better prediction accuracy [27].

Each LSTM is composed of input gate i_t , output gate O_t , forget gate f_t , and cell state memory unit C_t . i_t determines the information which needs to be transferred to the current state C_t according to the previous state h_{t-1} and the current input information. f_t determines what information from C_{t-1} is ignored and how C_{t-1} is updated to the current cell $\tilde{C}_t$. O_t determines how the current cell C_t affects the current hidden layer state h_t . LSTM output is

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \\ C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ h_t &= o_t * \tanh(C_t). \end{aligned} \quad (10)$$

Here, W_j and b_j , $j = f, i, C, o$ are the corresponding weight matrices and bias. σ is the sigmoid activation function.

The forward propagation LSTM reads the data sequentially and obtains the historical data information of the input sequence, and its output is

$$\vec{h}_t = \overrightarrow{\text{LSTM}}(x_t, \vec{h}_{t-1}). \quad (11)$$

The backward-propagation LSTM reads the future data information in a reverse order, and its output is

$$\overleftarrow{h}_t = \overleftarrow{\text{LSTM}}(x_t, \overleftarrow{h}_{t+1}). \quad (12)$$

The output of biLSTM is the concatenation of forward propagation and backward propagation results

$$h_t = \text{concat}(\vec{h}_t, \overleftarrow{h}_t). \quad (13)$$

The final output h_t of biLSTM gives the RUL prediction, i.e., $f_{\text{pre}} = h_t$. BiLSTM gives the point prediction result. Considering the cognitive uncertainty of the point prediction, the Bayesian deep learning strategy is introduced for biLSTM training to obtain the interval prediction.

C. Bayesian Deep Learning for Interval Prediction

The Bayesian theory is integrated into the BiLSTM model to transform the network parameter from fixed value into random variable. The probability distribution of parameter is derived to deal with the uncertainty modeling problem [28]. For the RUL interval prediction model biLSTM, its random model parameters are denoted as $\theta = [W_j, b_j]$, $j = i, f, C, o$ with prior distribution $p(\theta)$. Model output y^* 's conditional distribution is

$$p(y^* | x^*, X, Y) = \int p(\theta | X, Y) p(y^* | x^*, \theta) d\theta \quad (14)$$

where (X, Y) are the known sampling dataset and x^* is the real measured data. Under the Bayesian learning framework, the model parameter $\theta \sim N(\mu, \sigma^2)$ is random, so the prediction y^* is also random. The posterior distribution is

$$p(\theta | X, Y) = \frac{p(\theta)p(Y | X, \theta)}{p(Y | X)} \quad (15)$$

where $p(Y | X)$ represents the edge distribution of given dataset (X, Y) which is constant. $p(Y | X, \theta)$ stands for likelihood function.

The posterior probability $p(\theta | X, Y)$ is difficult to obtain due to the high-dimensional features of data and the complexity of network structure. $p(y^* | x^*, X, Y)$ is impossible to be directly estimated. VI is given to effectively solve this problem. Compared to expectation maximization (EM) and Monte Carlo (MC) methods, VI has several key advantages, such as improved computational efficiency, more reliable convergence, and greater stability. In particular, when dealing with large-scale data or complex models, EM and MC methods could cause the optimization process to become slow or even fail to converge. The VI algorithm constructs a variational distribution $q_\varphi(\theta)$ to close $p(\theta | X, Y)$ through continuous optimization. First, define the variational distribution as a parameterized probability distribution of φ ; Second, minimize the Kullback-Leibler (KL) divergence between $p(\theta | X, Y)$ and $q_\varphi(\theta)$

$$\begin{aligned} \text{KL}(q_\varphi(\theta) \| p(\theta | X, Y)) &= \int q_\varphi(\theta) \log \frac{q_\varphi(\theta)}{p(\theta | X, Y)} d\theta \\ &= \log p(Y | X) - \left(\int q_\varphi(\theta) \log \frac{p(Y | X, \theta) q_\varphi(\theta)}{p(\theta | X, Y)} \log d\theta \right) \\ &= \log p(Y | X) \\ &\quad - (E_{q_\varphi(\theta)}[\log p(Y | X, \theta)] - \text{KL}(q_\varphi(\theta) \| p(\theta))) \end{aligned} \quad (16)$$

This objective is divided into three terms: 1) $\log p(Y | X)$ is constant with respect to $q_\varphi(\theta)$; 2) $E_{q_\varphi(\theta)}[\log p(Y | X, \theta)]$ is the log-likelihood expectation aiming at the good interpretation of X and Y ; 3) the KL divergence term aims at the approximation of variational distribution $q_\varphi(\theta)$ and the posterior distribution $p(\theta)$. So problem (16) is transformed to the maximization of evidence lower bound (ELBO)

$$\begin{aligned} \text{ELBO} &= E_{q_\varphi(\theta)}[\log p(Y | X, \theta)] - \text{KL}(q_\varphi(\theta) \| p(\theta)) \\ &= \log p(Y | X) - \text{KL}(q_\varphi(\theta) \| p(\theta | X, Y)). \end{aligned} \quad (17)$$

Finally, a VI loss function is to minimize as follows:

$$\begin{aligned} L_{\text{VI}}(\varphi) &= - \sum_{i=1}^N \int q_\varphi(\theta) \log p(y_i | x_i, \theta) d\theta + \text{KL}(q_\varphi(\theta) \| p(\theta)). \end{aligned} \quad (18)$$

Usually we randomly select the small batch data $M \subset (X, Y)$ to simply the optimization of (18), denoted as the stochastic VI (SVI)

$$\begin{aligned} L_{\text{SVI}}(\varphi) &= - \frac{N}{M} \sum_{i \in S} \int q_\varphi(\theta) \log p(y_i | x_i, \theta) d\theta \\ &\quad + \text{KL}(q_\varphi(\theta) \| p(\theta)) \end{aligned} \quad (19)$$

where $S \subset \{X, Y\}$ represents a random index set of data size is M . At the same time, MC sampling method and parametric method are used to optimize the log-likelihood term in (18). Consider that the initial $q_\varphi(\theta)$ is a Gaussian distribution, and the parameters are updated by random sampling. So variational parameters $\varphi(\theta)$ is used to optimize the network parameter θ

$$\begin{aligned} q_\varphi(\theta) &= q_{\mu, \sigma}(\theta) = N(\mu, \sigma^2) \\ N(\mu, \sigma^2) &= \mu + \sigma \cdot N(0, I) \end{aligned} \quad (20)$$

where variational parameters φ is (μ, σ) . By the random sampling to approximate the log-likelihood expectation, (19) is further derived as follows:

$$\hat{L}_{\text{MC}}(\varphi) = - \frac{N}{M} \sum_{i \in S} \log p(y_i | x_i, \varphi, \mu, \sigma) + \text{KL}(q_\varphi(\theta) \| p(\theta)). \quad (21)$$

The model parameter $\hat{\theta}_t$ is obtained from the optimal variational distribution q_φ^* by MC sampling. The output distribution approximately is

$$\frac{1}{T} \sum_{t=1}^T p(y^* | x^*, \hat{\theta}_t) \xrightarrow{T \rightarrow \infty} \int p(y^* | x^*, \theta) q_\varphi^*(\theta) d\theta. \quad (22)$$

The predicted distribution is further obtained as

$$\begin{aligned} p(y^* | x^*, X, Y) &\approx \int p(y^* | x^*, \theta) q_\varphi^*(\theta) d\theta \\ &=: q_\varphi^*(y^* | x^*). \end{aligned} \quad (23)$$

D. RUL Interval Prediction of Aero Engine

When dealing with the data-driven RUL prediction problem of aero engine, the m -dimensional monitoring data

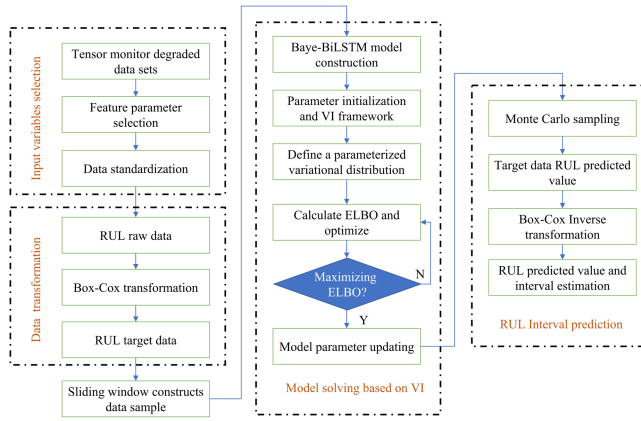


Fig. 3. Flowchart of RUL prediction algorithm. It outlines the main processing stages, including input variables selection, data transformation, VI-based model solving and interval prediction.

are sampled which describe the engine performance status $X \in R^{n \times m}$ and the corresponding residual life is $Y \in R^{n \times 1}$, where m and n are the number of independent monitoring variables and the samplings. The training dataset is defined as $D = (X, Y)$. The prediction y from the monitoring state variable x is given by the BCT-Bayesian-biLSTM network, i.e., $y = f_{\text{pre}}(x, \theta) + \varepsilon$. Here, θ is the model parameters, and the term ε is generally described by Gaussian white noise. The prediction function f_{pre} fully considers the influence of uncertainty factors, and the output distribution is (23).

Based on the above analysis of BCT-Bayesian-biLSTM, the RUL prediction of aeroengine is given in Fig. 3.

Step 1: Data preprocessing and data analysis. The multi-dimensional sensor monitoring data are analyzed in-depth. Generally, the original sensor data includes null value and outlier which should be deleted. Correlation analysis and mechanism analysis are often used to determine the independent variables as the input for RUL prediction. The sliding window method is introduced to construct the training samples.

Step 2: Data transformation. BCT is used to transform the label data RUL to the target data to guarantee its distribution matching with the assumed one.

Step 3: Bayesian-biLSTM network training. Construct a biLSTM network for RUL prediction and initialize the prior probability distribution of model parameter. The Bayesian deep learning strategy is used for network training based on SVI framework. The parametric variational distribution q_{ϕ} is found to approximate the posterior distribution by minimizing the KL divergence between two distributions and the log-likelihood expectation (18).

Step 4: RUL prediction. Based on the trained network, the RUL of aeroengine is predicted under the new monitoring data. The optimal variational distribution is found based on MC sampling. The obtained prediction value is in the RUL target data space which should be restored into the original data space by inverse BCT. Finally, the prediction value and its interval estimation are obtained.

To evaluate the prediction performance, two evaluation index are used, score function and root mean square error

TABLE I
Description Subset of C-MAPSS Datasets

Subset	Num. of fault modes	Num. of operation condition	Training sequences	Testing sequences
FD001	1	1	100	100
FD002	1	6	260	259
FD003	2	1	100	100
FD004	2	6	249	248

TABLE II
Description of 26 Columns in C-MAPSS Datasets

Column Number	Description
1	Unit number
2	Time in cycles
3	Operation setting 1
4	Operation setting 2
5	Operation setting 3
6	Sensor S_1
7	Sensor S_2
...	...
26	Sensor S_{21}

(RMSE). The smaller the RMSE and score values, the better the prediction performance of the model. RMSE represents the deviation between the true value y_i and its prediction $\hat{y}_i$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (24)$$

where N is the number of samples. Score indicates the score value of RUL prediction results

$$\text{Score} = \begin{cases} \sum_{i=1}^N \exp\left(-\frac{\hat{y}_i - y_i}{13}\right) - 1, & \hat{y}_i - y_i < 0 \\ \sum_{i=1}^N \exp\left(\frac{\hat{y}_i - y_i}{10}\right) - 1, & \hat{y}_i - y_i \geq 0. \end{cases} \quad (25)$$

Compared with the advance prediction, delayed prediction may lead to more serious consequences. The scoring function could punish the delayed prediction more severely. The lower the score, the higher the accuracy of the prediction.

IV. EXPERIMENT ANALYSIS

This article uses the C-MAPSS degradation dataset of the turbofan aero engine published by NASA [29]. The C-MAPSS dataset records many sensor information for the entire engine life cycle, i.e., from the normal operation to failure. The dataset is divided into four subsets (FD001–FD004) under different working conditions, as shown in Table I. Each subset is composed of training set, test set, and related RUL dataset. The training set and the test set store the data with 26 columns detailed in Table II. It includes the engine number, the cycle period, a list of the operating condition settings, such as height, Mach number, and throttle rotation angle, and the 21-D sensor monitoring data such as speed, gas flow, pressure, etc. In this article, the FD001 dataset of single operating condition is used to verify the method, and the other subsets are used for model training. The detailed experimental environment is shown in Table III.

TABLE III
Experimental Environment Setting

Name	Version
Hardware	Intel i7 CPU, 16GB RAM, N-RTX 3060
Software	Pycharm2023.2.7
Programming language	Python3.8.19
Model framework	Pytorch2.2.2, Torchvision0.17.2
Train Dataset	0.8*Train-FD001
Val Dataset	0.2*Train-FD001
Test Dataset	Test-FD001

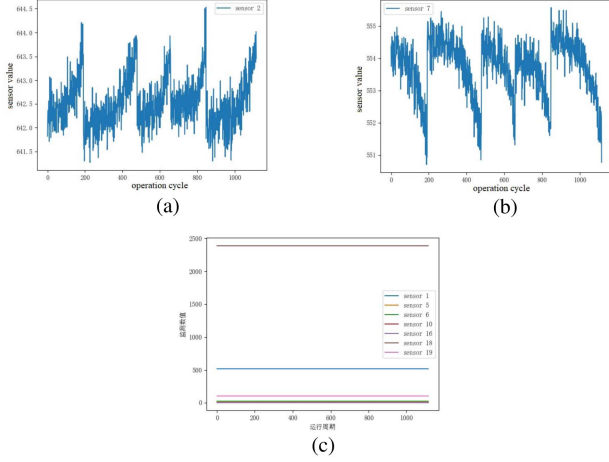


Fig. 4. Representative data analysis. (a) Upward trend indicating progressive system degradation. (b) Downward trend reflecting declining performance. (c) Constant signal suggesting minimal correlation with degradation behavior. (a) Upward: Sensor2. (b) Downward: Sensor7. (c) Constant: Sensor1,5,6,10,16,18,19.

A. Data Description and Preprocess

The multidimensional data monitored by sensors can reflect the degradation law of the engine. The 21-D sensors data in the FD001 subset is analyzed. However, not all sensor data have rich information that contributes to the RUL estimation. For example, measurements from several sensors are constants throughout the life cycle [21]. It is found that the monitoring variables of 21 sensors are divided into three categories, seven variables are steady, ten variables show the upward trend, and four variables show the downward trend with the increase of the operation cycle of the engine, exemplified in Fig. 4. These examples highlight the diverse characteristics of sensor outputs and provide guidance for effective sensor selection in prognostic modeling. Variables with upward trend reflect increasing load or decreasing efficiency. They serve as important indicators for modeling the degradation process. In contrast, downward trend of variables signify the performance deterioration and directly indicate the decline in equipment health. The variables with constant values are insensitive to degradation and contribute little to prediction. So they are excluded to reduce computational cost and improve efficiency. Therefore, 14 monitoring variables showing upward and downward trends are selected as inputs for RUL prediction.

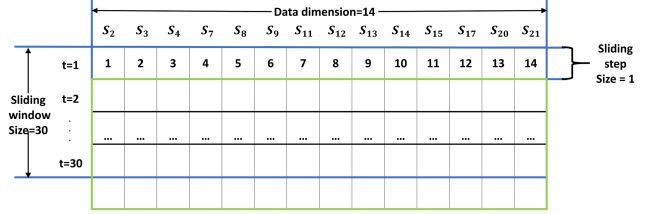


Fig. 5. Sliding window strategy. A fixed-length window (size 30) with stride 1 is applied to the selected 14 sensor signals. The generated overlapping subsequences rich local temporal characteristics which are good for RUL prediction.

The maximum-minimum normalization is first used to preprocess the sensor monitoring data

$$x_{ij}^* = \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}} \quad (26)$$

where x_{ij} is the original data from sensor j at the time i , $x_j^{\max}$ and $x_j^{\min}$ are the maximum and minimum value of column j , and x_{ij}^* is the normalized value.

To deal with the overfitting issue, several strategies are employed. Bayesian inference is adopted to regularize the model by approximating the posterior distribution of network parameters, while also providing uncertainty quantification. Dropout is applied to randomly deactivate neurons and enhance the generalization during the training. Early stopping is used based on validation loss to prevent excessive training. In addition, data normalization and label segmentation are conducted to reduce input variance and stabilize the learning process.

Here, we use the sliding window to reconstruct the input. Suppose that a sample intercepted by the sliding window is $[X_r, RUL_r]$, where X_r are the sensor monitoring data between time $[t_r, t_{r+l}]$. RUL_r is the value of RUL at the intercepted time. Reconstructed samples contain more information to the maximum extent, and has the obvious advantage of maintaining local dependence. The construction method is shown in Fig. 5. Based on the above sliding window construction training sample method to process the FD001 dataset, sensor monitoring data feature dimension 14 as the sliding window width. The step size and window size are 1 and 30, respectively.

In order to reduce the interference of engine normal state features on degraded state features, RUL label are usually processed in segments. A fixed RUL value T is set for the early normal states. When the RUL value is greater than or equal to the fixed T , it means that the engine is in normal state and the RUL label is set as T . When the RUL value is less than T , the engine begins to degrade. The RUL label is consistent with the true value. The segmented representation of RUL data is

$$Y_{\text{label}} = \begin{cases} T, & Y \geq T \\ Y, & Y < T. \end{cases} \quad (27)$$

The selection of T is important. Too small T results in prematurely entering the degradation modeling stage, so it is difficult to capture features of the healthy state. Conversely,

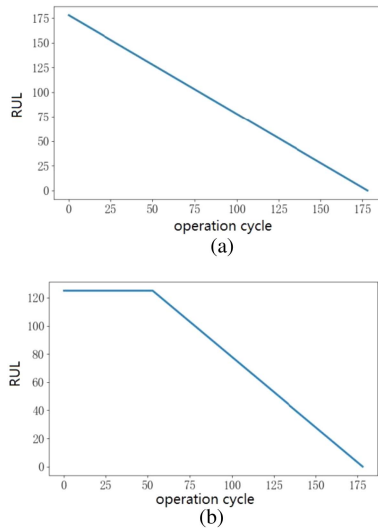


Fig. 6. RUL labeling strategy. (a) Raw value decrease linearly until failure. (b) Segmented label with a cap at 125. It highlights the later degradation patterns, so reduces variance and stabilizes training. (a) Raw RUL. (b) Segmented label.

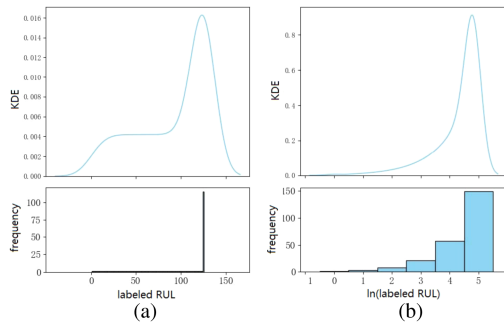


Fig. 7. Distribution of labeled RUL. (a) Original RUL distribution before BCT shows irregularities. (b) RUL distribution after BCT exhibits more structured pattern aligned with the target one. (a) Before BCT. (b) After BCT.

an excessively large T results in an insufficient proportion of degradation samples, limiting the model's ability to learn key degradation features. The fixed value of T is closely determined by the degradation state of aeroengine. In this study, based on the life cycle distribution of different subsets in the CMAPSS dataset and considering the average number of cycles from the onset of degradation to failure, a threshold of running cycles $T \in [120, 130]$ is appropriate. Therefore, the maximum cycle $T = 125$ is selected to effectively separate normal operational features from degradation features. The reconstructed RUL segmented label is given in Fig. 6.

Fig. 7 shows the kernel density estimation of data distribution and the frequency histogram of label data before and after BCT. It can be seen that original RUL does not obey normal distribution, so it is necessary to transform it to assure its distribution closer to the normal hypothesis. Here, the BCT parameter $\lambda = 0$ which is the natural logarithm transformation. It helps to reduce the left or right data bias and make it more close to the normal distribution. It is found that BCT can significantly reduce the right-skewness of data.

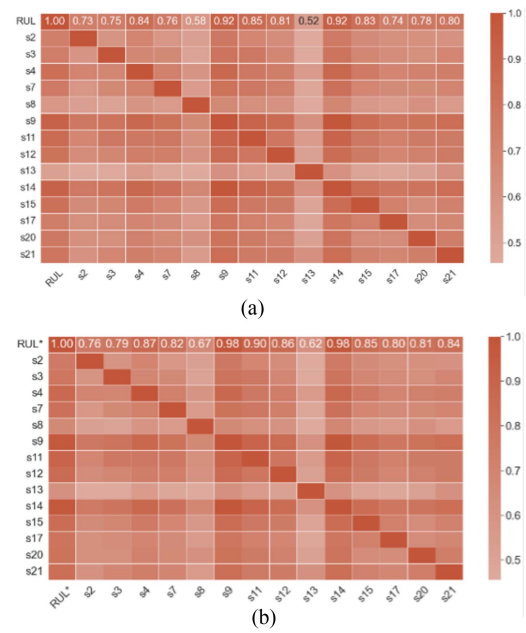


Fig. 8. Correlation analysis between monitoring variables and degradation performance. (a) Before BCT. (b) After BCT. For example, correlation between variable S_2 and RUL increase from 0.73 to 0.76 after BCT. The stronger the correlation is, the higher the prediction accuracy will be. (a) Before BCT. (b) After BCT.

TABLE IV
Model Parameter Setting

Parameter	Value
Learning rate	0.01
Epochs	500
Batch size	256
Hidden layer	BiLSTM layer
Number of neurons	64
Loss function	Trace_ELBO
Initial prior distribution	$N(0, 1)$

BCT transforms the raw RUL data into the target data, expressed as RUL*. The correlations between RUL* and 14 monitoring variables are analyzed. The 14 monitoring variables are sampled from fan and compressors to characterize the degradation trend of the engine performance. The correlations were drawn as a thermal map, as shown in Fig. 8. It is found that the BCT increases the correlation between labeled RUL and sensor monitoring data which is helpful for the model learning.

B. Prediction Result and Analysis

The model parameters of BCT-Bayes-BiLSTM are given in Table IV, and the loss function of the training process is shown in Fig. 9(a). The model loss almost no longer declines after 200 epoch. The trained model is tested in the FD001 subset. The testing result show that the RMSE index of RUL prediction is 14.9188, and the score index is 314.7840. Fig. 9(b) gives the histogram of prediction error with range $[-45, 35]$. Most of the errors exist in the range of $[-5, 5]$. A few error locates in the range of $[-45, -30]$ and $[30, 35]$. It shows that the prediction error is controllable in general.

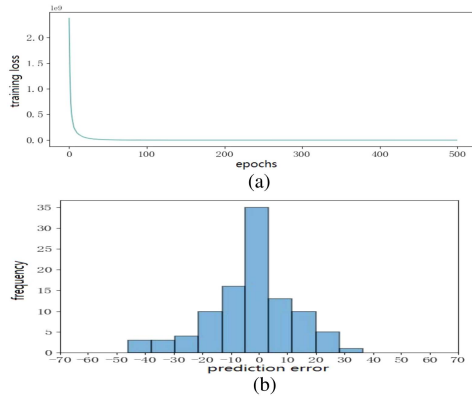


Fig. 9. Training and test results. (a) Training loss converges within 100 epochs, indicating stable optimization. (b) Histogram of test errors shows most predictions are close to zero, with smaller peaks at ± 10 and ± 20 , reflecting accurate and balanced performance. (a) Training loss. (b) Testing error histogram.

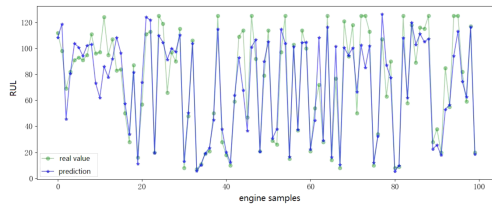


Fig. 10. RUL prediction results for 100 test engines. The prediction closely match the true value, demonstrating robust performance under diverse degradation patterns.

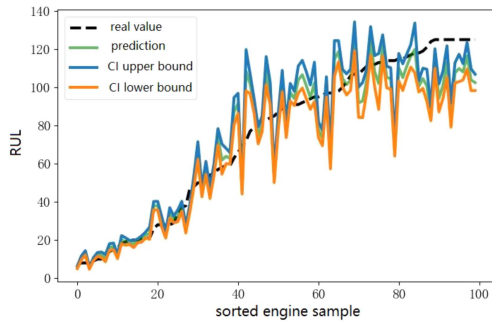


Fig. 11. Sorted RUL predictions with confidence intervals. Predicted RUL and uncertainty bounds align well with ground truth, highlighting both accuracy and reliable uncertainty estimation.

Fig. 10 gives the RUL prediction of 100 engines. It can be seen that there are more engines with advanced prediction. When the true value of engine RUL is smaller, the predicted value is closer to the true value. The 100 engine samples in the FD001 subset have different running cycles. Fig. 11 displays the prediction results sorted in ascending order and the corresponding confidence intervals. It is found that the prediction confidence interval is relatively narrow when the true RUL value is small, and the prediction is closer to the true value. On the contrary, the confidence interval is wider when the RUL true value is large, and the prediction deviates from the true value to some extent. Furthermore, the prediction accuracy is related to the information amount of sensor monitoring data. For

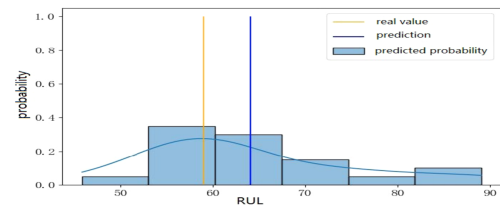


Fig. 12. Predicted RUL distribution for engine No. 43 in the test set. The yellow line indicates the real value, the blue line is the prediction, and the histogram represents the predicted probability distribution.

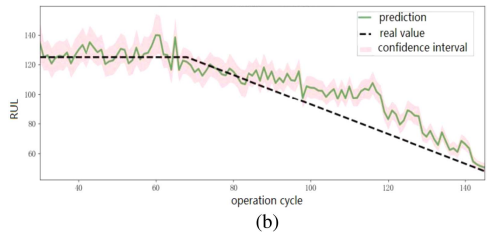
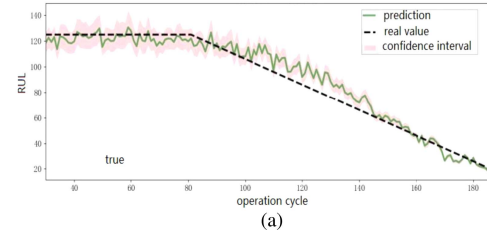


Fig. 13. RUL prediction results for engine no. 24 and engine no. 32. Black dashed line indicates the real value, and green line is the predictions, and the shaded pink area means its confidence interval. (a) Engine no. 24. (b) Engine no. 32.

the engine whose true RUL value is greater than 125, the engine is operated in normal condition. Since there is no obvious degradation trend which are shown from the sensor monitoring data, it results the model in a lower prediction accuracy. The engine who has smaller RUL true value, the richer information is shown from the monitoring data, and the higher prediction accuracy is obtained.

The output of BCT-Bayes-BiLSTM model is the distribution of RUL prediction. Multiple values of the residual life prediction are obtained by sampling from the prediction model, then the mean and variance are calculated to give its probability distribution. Fig. 12 shows the prediction probability distribution of engine No. 43 with 20 sampling times. The RUL prediction is near the truth and the high prediction accuracy also implies the prediction reliability.

The statistical analysis is performed in depth with 99% confidence interval $[(\mu \pm 3\sigma)]$ principle. Fig. 13 displays the interval prediction for engine No. 24 and 32, respectively. The black dotted line denotes the real RUL, the green solid line denotes the average prediction with 20 times of sampling, and the pink domain is the confidence interval. It is found that the prediction mean value μ has the same change trend as the real value during the whole engine operation cycle, and the 99% confidence interval covers most of the real value. In addition, when the operation cycle is more close to its maximum cycle, the residual life is

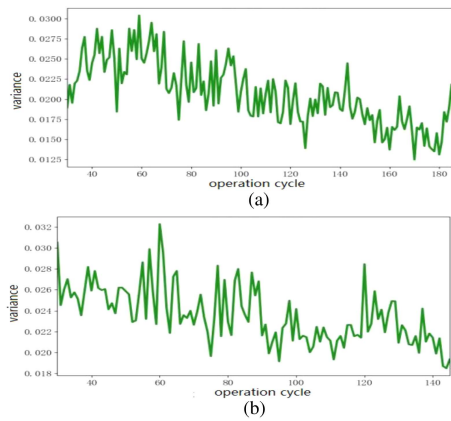


Fig. 14. Prediction variance. It represents the modeling uncertainty over the operation cycles in which (a) engine no. 24 and (b) engine no. 32 (b).

smaller and the prediction is more consistent with its real value. The mean of multiple prediction results obtained by the sampling model can be used as the point prediction value which fits the real value well. The prediction uncertainty of RUL can be expressed by the confidence interval deviating from the mean (such as variance or standard deviation). Compared with the traditional point estimation, the proposed model gives the lower and upper bounds of RUL prediction, and provides more comprehensive and interpretable information for subsequent maintenance decision.

The change of variance for engine No. 24 and 32 is used for evaluate the prediction uncertainty, as shown in Fig. 14. On the whole, the prediction uncertainty shows a decreasing trend with the increase of the operation cycle, specifically consistent with Fig. 13. The width of confidence interval in the early operation is slightly larger than the later operation. The prediction model in the early stage has higher uncertainty and larger prediction error than in the later stage. With the increasing operation cycle, more degradation features are extracted, and the prediction performance at accuracy (mean) and uncertainty (variance) are improved, i.e., the mean is more close the real RUL value, the variance is increasingly smaller, and the confidence interval continuously narrows.

C. Comparative Study Result and Analysis

To further validate the superiority of the proposed method, comparative experiments are conducted. Here, the traditional predictive models (BP, LSTM, BNN, Bayesian-LSTM, Bayesian-BiLSTM) and the state-of-the-art (SOA) predictive models (GCUTransformer [21], SSA-KELM [18], GGCN [20], GCN-SAGpool [19]) are selected. All experiments were performed on the C-MAPSS FD001 dataset to ensure fairness and consistency. The labels, data, training epochs, and evaluation metrics were kept consistent with those used in the BCT-Bayesian-BiLSTM model. Since the BCT-Bayesian-BiLSTM model converges by the 100th epoch, the comparative experiments were also run for 100 epochs. The evaluation metrics used for comparison were RMSE (24) and score (25). RMSE measures the average deviation between predicted and actual values. It is

TABLE V
Comparison of Model Performance

No	Experimental method	RMSE	Score
1	GCN-SAGpool	15.34	1264.5
2	SSA-KELM	14.2224	351.1711
3	GGCN	12.8692	362.8256
4	GCU-Transformer	11.6332	226.9405
5	BP	21.0668	1744.5463
6	LSTM	20.7695	1494.6795
7	BNN	19.2427	1112.9111
8	Bayesian-LSTM	18.3253	792.4359
9	Bayesian-BiLSTM	15.2233	392.2314
10	BCT-Bayesian-BiLSTM	14.9188	314.7840

widely used for evaluating regression models but is highly sensitive to large errors, which may not fully reflect the model's performance. On the other hand, the score index provides a more flexible evaluation by using an exponential function to weigh the errors. It allows to better capture the model's performance across different error ranges.

The comparison results are summarized in Table V. Compared with traditional methods (BP, LSTM, BNN, Bayesian-LSTM, Bayesian-BiLSTM), the proposed method shows a clear advantage in both RMSE and score. Compared with the other SOA models (SSA-KELM, GGCN, GCN-SAGpool), the proposed model also demonstrates the similar RMSE values but significantly outperforms them in score metric. Although the GCU-transformer model, shows superior performance at the cost of a significant increase in model complexity. So it is not a good choice for the real industrial application.

The comparison result shows that the proposed method has the best interval prediction accuracy. By introducing the Bayesian inference into the biLSTM network, we can predict the accuracy RUL value and its confidence interval. The bidirectional design can extract more sequential features between monitoring data and the degradation property. The cognitive uncertainty is quantified which enhances the robustness of network model. Moreover, the BCT mechanism introduced here can reduce the impact of uncertainty on the prediction accuracy, so the performance of prediction model is further improved.

V. CONCLUSION

The distribution of industrial data in different operating environments shows diversity and the inconsistency with the general assumption required by the deep learning method. To address this challenge, an aero engine RUL prediction method based on BCT-Bayes-BiLSTM is proposed in this article. The data transformation mechanism based on BCT is used to make the data distribution match with the assumed one. The biLSTM under the Bayesian framework efficiently captures the temporal correlation hidden in process data which helps to improve the prediction accuracy. At the same time considering the cognitive uncertainty naturally existing in any data learning model, the interval prediction of RUL is complemented by the Bayesian deep learning strategy. The accurate RUL value and its confidence interval are predicted simultaneously. Experiment results on CMAPSS

data show that the highest prediction accuracy and narrowest uncertainty interval are obtained. It reflects the reliability of prediction result.

There are several constraints that may affect the real implementation of the proposed model, including the quality of collected industrial data, the complexity of the deep learning model, and the limitations of edge computing devices. In addition, real-time inference delays caused by computational resource limitations and data transmission issues also need to be carefully considered. Future research should focus on improving data quality, optimizing the model for low-latency inference, and developing lightweight models for deployment on edge devices.

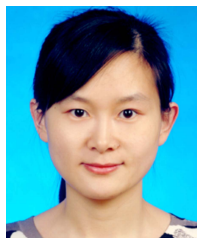
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